

Project Title: Cloud-Based Website Hosting Using Amazon Web Services (AWS)

Abstract

Cloud computing provides scalable, reliable, and cost-effective solutions for hosting websites and web applications. Traditional hosting requires organizations to purchase and maintain physical servers, manage storage, install software, perform backups, and manually handle increases in traffic. These activities increase operational cost and complexity. This project demonstrates how Amazon Web Services (AWS) can be used to host a website using managed cloud services. AWS services such as EC2, S3, Elastic Load Balancer, Auto Scaling, Route 53, CloudFront, IAM, and CloudWatch provide secure, reliable, and highly available hosting. The purpose of this project is to understand cloud deployment, reduce infrastructure cost, improve performance, and gain practical knowledge of cloud technologies. The expected outcome is a secure, scalable, and efficient website deployment that can handle changing traffic with minimal downtime while following a pay-as-you-go pricing model.

Introduction

The growth of internet services has increased the demand for reliable hosting. Traditional infrastructure requires large investments and regular maintenance. Cloud computing allows organizations to access computing resources over the internet whenever needed. AWS offers a wide range of services that simplify deployment, storage, networking, monitoring, and security. Using cloud hosting helps businesses focus on application development instead of hardware management.

Problem Statement

Traditional website hosting suffers from high hardware costs, limited scalability, manual maintenance, downtime during failures, and complex disaster recovery. Businesses need a flexible and secure solution that scales automatically and reduces operational effort.

Technology Used

Amazon EC2, Amazon S3, Elastic Load Balancer, Auto Scaling, Amazon Route 53, Amazon CloudFront, IAM, CloudWatch, HTML, CSS, and JavaScript.

Purpose of the Project

To deploy websites on AWS, reduce maintenance costs, improve availability, ensure security, and understand practical cloud computing concepts.

Expected Outcome

The website will achieve high availability, automatic scaling, secure access, faster content delivery, and lower infrastructure costs.

Key Points

- Cloud hosting
- Auto Scaling
- Load balancing
- Secure access using IAM
- Monitoring with CloudWatch
- Fast delivery using CloudFront
- Reliable storage using S3

Advantages

Lower cost, scalability, flexibility, reliability, high performance, automatic backup, easy maintenance, and global accessibility.

Limitations

Internet dependency, learning curve, service costs if resources are unmanaged, and dependence on the cloud provider.

Components in the Diagram

User: accesses the website. Route 53: directs requests. Load Balancer: distributes traffic. EC2: runs the application. Auto Scaling: adjusts server count. S3: stores files. CloudFront: speeds up delivery. IAM: manages permissions. CloudWatch: monitors performance.

Future Scope

Integrate AI-based monitoring, serverless computing with AWS Lambda, Docker and Kubernetes deployment, DevOps CI/CD pipelines, and advanced security services.

Conclusion

Cloud-based website hosting using AWS is an efficient, scalable, secure, and economical solution for modern web applications. It reduces maintenance, improves performance, and prepares students for real-world cloud computing environments. **Project Title:** Cloud-Based Website Hosting Using Amazon Web Services (AWS)

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